PhySci/MiMG/CaSB M178

Homework 3

Due: 10/21/25 at 2:00PM PDT

Notes: This homework involves performing simulations of the caspase cascade we've been discussing in the last two class meetings. On the same page on BruinLearn where you obtained this document, you will also find a file called "HW3_template.ipynb" that contains a template Jupyter notebook that you can use as a starting point to complete the questions below. Please modify this notebook and use it as the starting point for answering the following problems.

To submit your homework, please answer the questions below. Note that you will have to **paste in several graphs** that you generate using the Jupyter notebook. After completing the questions, **save this document as a PDF** and upload it to Gradescope. **You must also upload the Jupyter notebook to Bruinlearn**. Finally, do not forget to **MARK YOUR QUESTIONS** when uploading to Gradescope.

Part 1: Saturating Caspase Model

In class, we talked about the caspase cascade. We initially considered a simple model in which the amount of caspase 8 in the model was fixed. Caspase 8 cleaves pro-caspase 3 to generate the active caspase 3 species. In this problem we will extend the model to include the saturation of caspase 8. To do this, we will need to model the formation of the complex of caspase 8 and pro-caspase 3.

$$C8 + proC3 \xrightarrow{k_a} C8proC3$$
$$C8proC3 \xrightarrow{k_d} C8 + proC3$$

As in the model from class we still have the conversion of pro-caspase 3 to caspase 3 via caspase 8, as well as the synthesis of pro-caspase 3, and the degradation of pro-caspase and caspase 3:

$$C8proC3 \xrightarrow{k_c} C8 + C3$$
$$Q \rightarrow proC3$$
$$proC3 \xrightarrow{\delta}$$
$$C3 \xrightarrow{\delta}$$
$$C8proC3 \xrightarrow{\delta} C8$$

We do not model the activation, synthesis, nor degradation of caspase 8 here.

1. (5 points) In the section titled “initial caspase model,” write down the change equations for this new caspase model including saturation. Paste a screenshot of your code here:

```
# Initial caspase model
def caspase_model(states,t,k_a,k_d,k_c,delta,Q):
    # Our "states" list has the state variables in the following order:
    # C8, proC3, C8proC3, C3
    # k_a,k_d,k_c,_delta,and Q are the parameters of the model as described in the homework document

    C8 = states[0]
    proC3 = states[1]
    C8proC3 = states[2]
    C3 = states[3]

    ### implement the change equations to reflect the new model
    C8_prime = -k_a*C8*proC3 + k_d*C8proC3 + k_c*C8proC3 + delta*C8proC3
    proC3_prime = -k_a*C8*proC3 + k_d*C8proC3 + Q - delta*proC3
    C8proC3_prime = k_a*C8*proC3 - k_d*C8proC3 - k_c*C8proC3 - delta*C8proC3
    C3_prime = k_c*C8proC3 - delta*C3
    ###

    statesprime = [C8_prime, proC3_prime,C8proC3_prime, C3_prime]
    return statesprime

###
```

0.0s

In the next section of code, “INITIAL CONCENTRATIONS & PARAMETERS”, the values we will use for simulation are set, except the value of Q, the synthesis of rate of pro-caspase 3. Imagine there was no caspase 8 in the cell, what must the value of Q be such that pro-caspase 3 synthesis balanced its degradation (i.e. there is no change in the concentration of pro-caspase 3 over time)?

2. (3 points) Enter your expression for the value of Q in the code and paste a screenshot of it below.

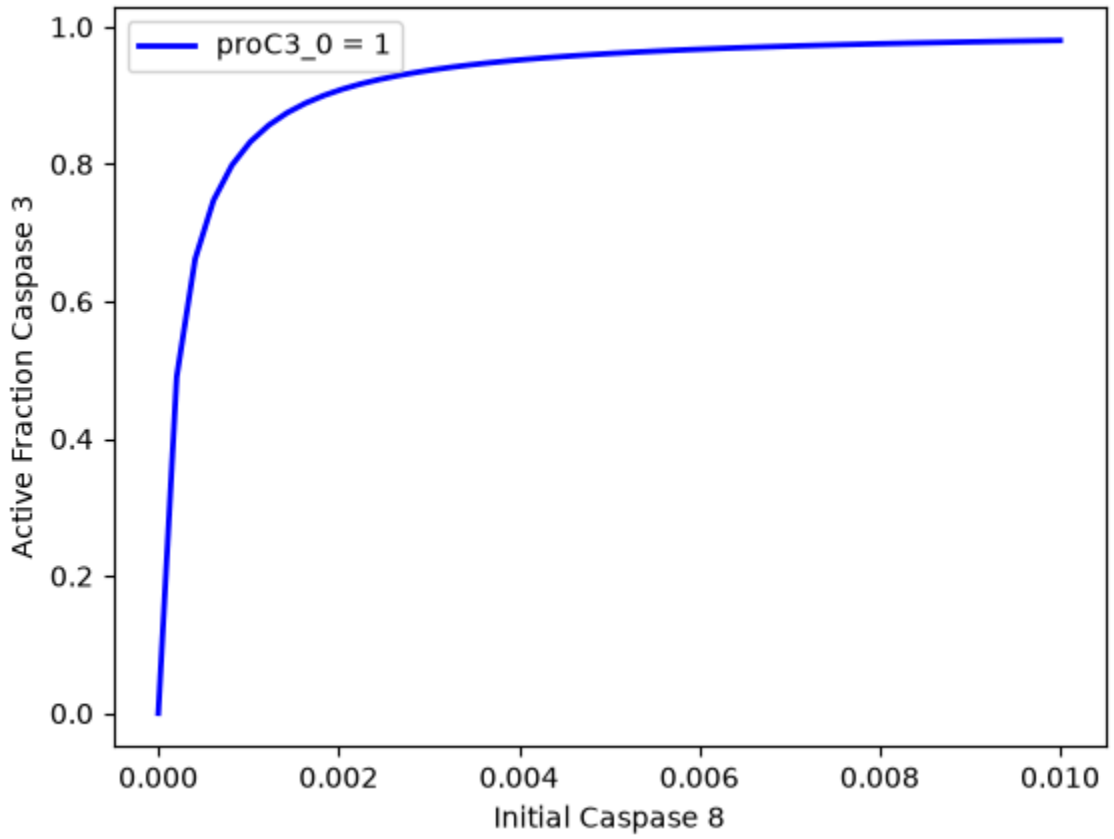
```
### enter your expression for the value of Q (synthesis rate of proC3)
Q = delta * proC3_0
```

@ proC3_0 = 1, q = 0.0002 * 1 = 0.0002

Finally in the section of code called “STEADY-STATE responses of initial caspase model”, we will vary the initial concentration of C8 and plot the resulting steady state concentration of C3 divided by the initial concentration of proC3. Note

that the current initial concentration of proC3 is set to 1 and the initial concentration of C8 will be varied from 0 to 0.01.

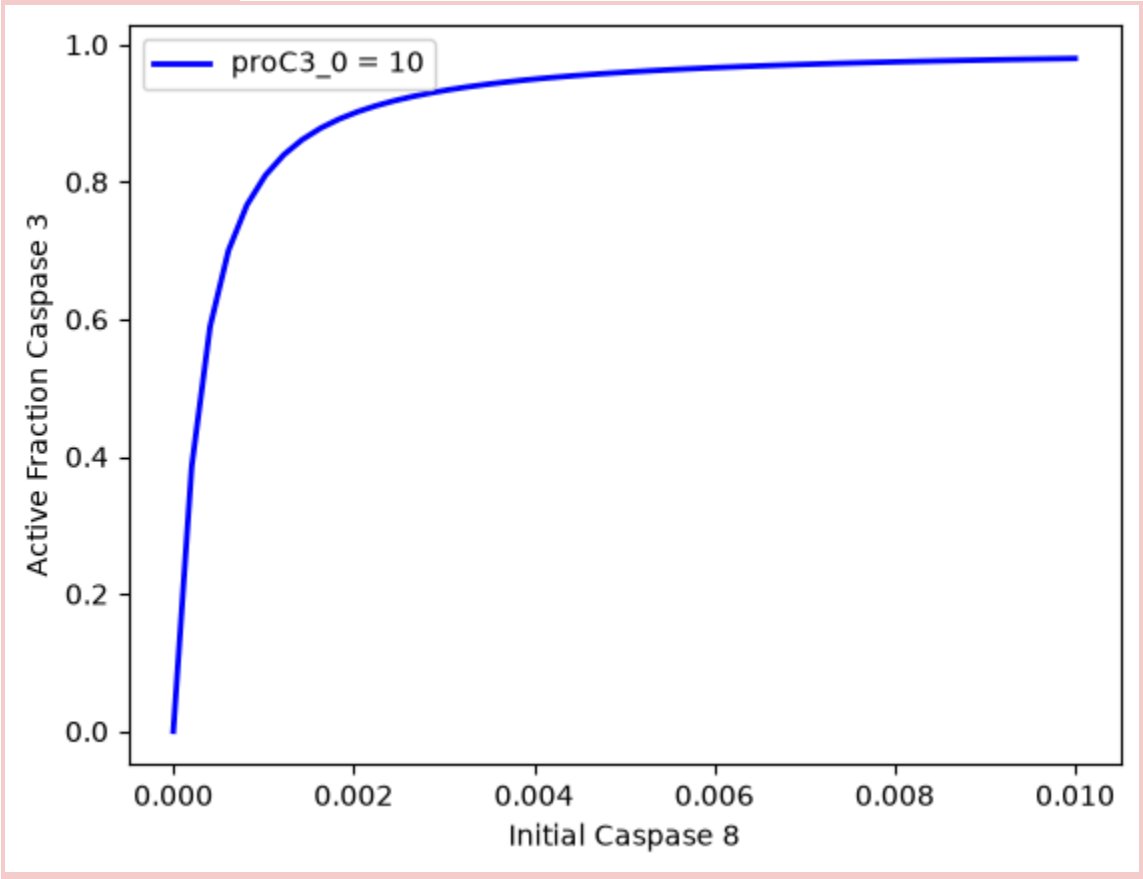
3. (3 points) Run this section of code and paste the resulting graph below.



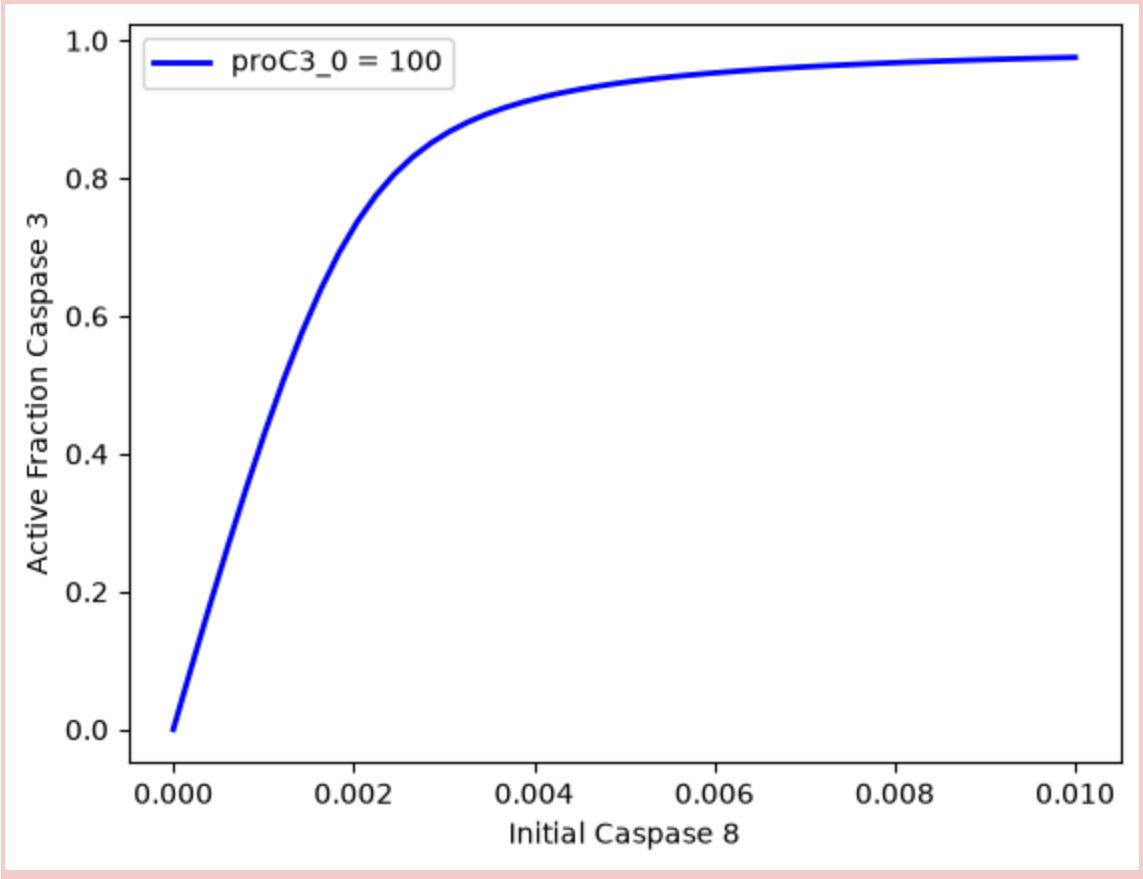
In HW 1, we explored the effect of saturation on the steady state amount of modified substrate as a function of initial kinase concentration by changing the initial concentration of substrate.

4. (6 points) Change the initial concentration of proC3 in this caspase model to explore the effect of saturation in this system. In the past two homework assignments, we've chosen the new parameters for you. Here, we'd like you to choose parameters to try. Increase the initial concentration of proC3 to the point where caspase C8 is saturated. Try two different values to see what happens. Rerun the code and paste your resulting graphs below. **Make sure to indicate what concentration you used** on your graphs. Remember the units of concentration are M!

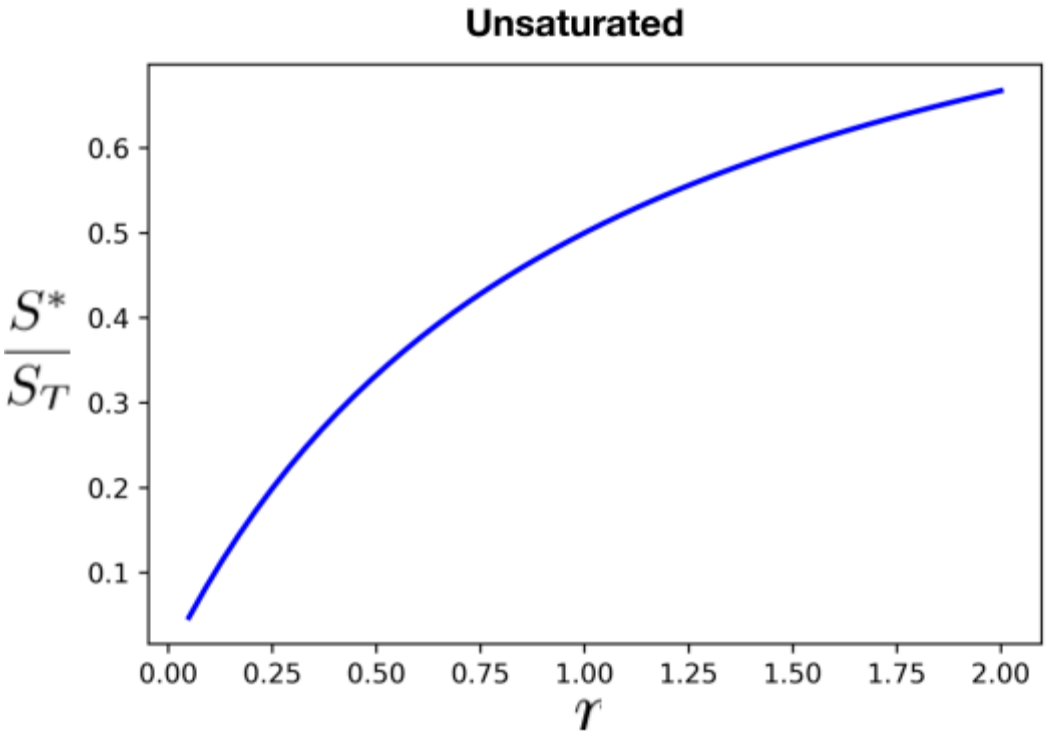
@ proC3_0 = 10:

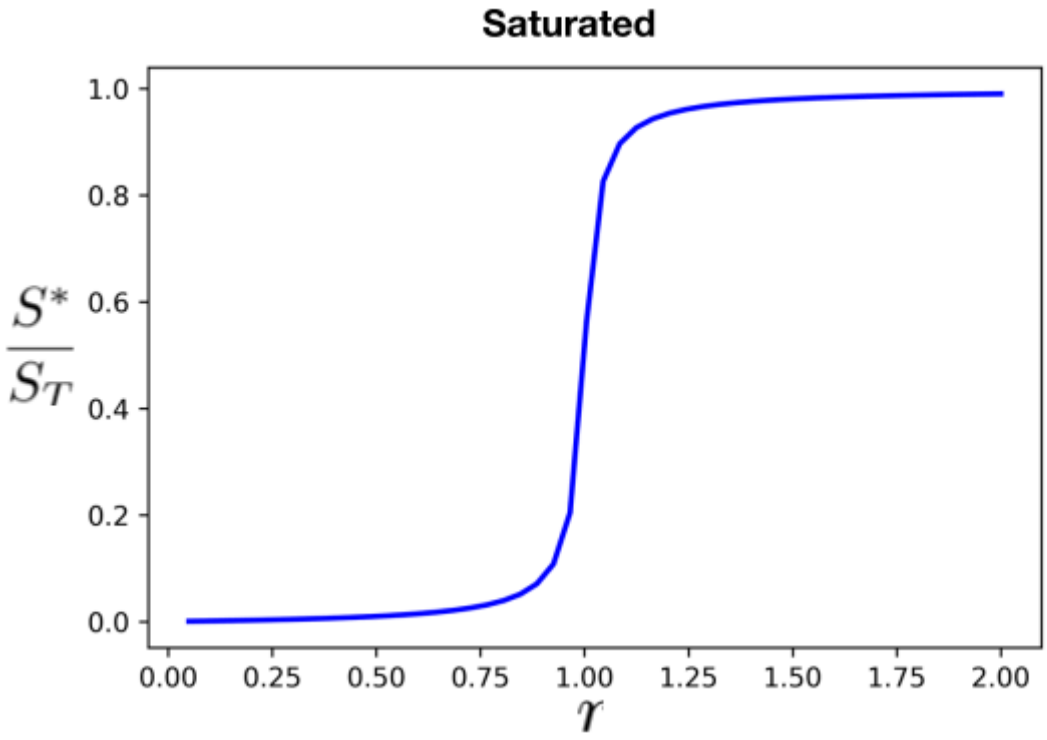


@ proC3_0 = 100:


proC3_0 = 100
Active Fraction Caspase 3
Initial Caspase 8
1.0
0.8
0.6
0.4
0.2
0.0
0.000
0.002
0.004
0.006
0.008
0.010

As a reminder, here is the result that we got for the simple Kinase-Phosphatase PTM cycle in HW1:





5. (5 points) How do your results (the graphs from question #4 above) compare to the above results? Do you see the same levels of ultrasensitivity in this model?

There are some clear differences between this caspase model and the saturated kinase phosphate model in HW1. When the initial procaspase 3 concentration increases from 10 to 100, the response becomes more gradual rather than switch-like. HW1s model made a sharp transition in substrate phosphorylation, exhibiting ultrasensitivity behavior. With this new model, increasing substrate concentration does not produce the same effect since caspace 3 activation is irreversible, and there are no opposing enzymes to reverse the reaction. Hence, this model does not show the same degree of ultrasensitivity as the saturated kinase phosphate model did.

Part 2: FLIP Caspase Model

Next, we will implement a version of the caspase model that includes a receptor activating pro-caspase 8, FLIP binding at the receptor, and positive feedback by caspase 3 in the section “simple caspase model with feedback and FLIP”.

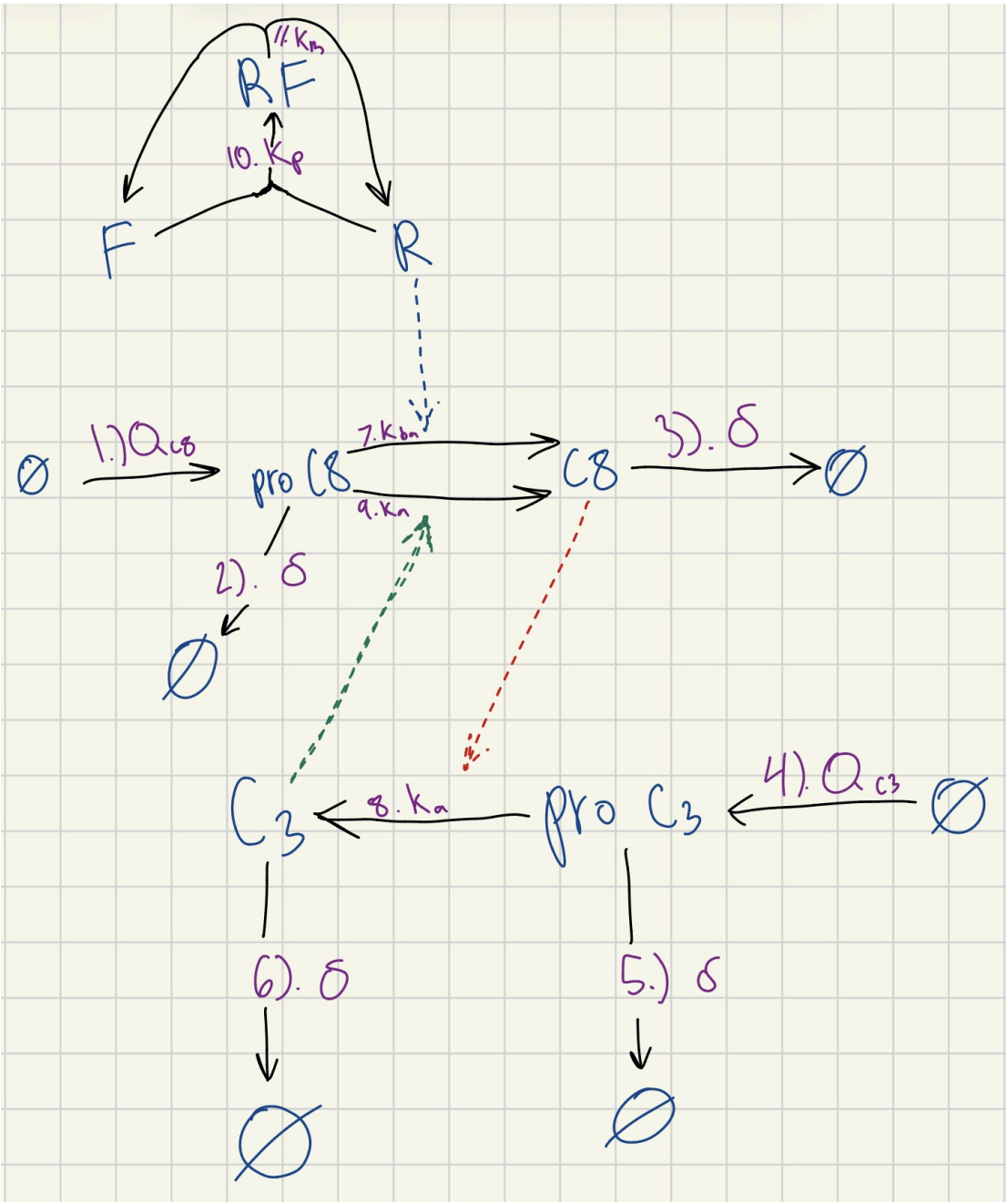
In this model we will have the following species: proC8, C8, proC3, C3, R (receptor), F (FLIP), and RF (receptor-FLIP complex).

Here are the assumptions we want to make:

a) proC8 and proC3 are synthesized at rate QC8 and QC3 respectively (zeroth order reaction).
b) proC8, C8, proC3, and C3 are all degraded at rate delta (first order reaction)
c) proC8 reacts with R at a rate k_ba and generates C8 (second order reaction)
d) proC3 reacts with C8 at rate k_a and generates C3 (second order reaction)
e) C3 reacts with proC8 at rate k_a and generates C8 (second order reaction)
f) F binds R at rate kp to generate RF (second order reaction)
g) RF dissociates into F and R at rate km (first order reaction)

Note that, when R is bound to F, it *cannot* react with proC8.

6. (8 points) Draw a reaction network diagram for this system below using the conventions we've developed in class: dashed arrows for catalysis and solid arrows for transformations. As in lecture, synthesis is represented by having an arrow go from nothing into the synthesized species, and binding reactions are represented by writing out the reaction explicitly. You should have a total of 7 species and 11 solid arrows (remember FLIP binding is reversible!). **Number each of your arrows 1-11.**



7. (4 points) Write the chemical reactions for this model below. Remember to label each reaction with the **corresponding number** from your network diagram above. Also, remember to include the rate constants corresponding to each reaction!

1.) $\varnothing \xrightarrow{Q_{C8}} proC8$

2.) $proC8 \xrightarrow{\delta} \varnothing$

3.) $C8 \xrightarrow{\delta} \varnothing$

4.) $\varnothing \xrightarrow{Q_{C3}} proC3$

5.) $proC3 \xrightarrow{\delta} \varnothing$

6.) $C3 \xrightarrow{\delta} \varnothing$

7.) $proC8 + R \xrightarrow{k_{ba}} C8 + R$

8.) $proC3 + C8 \xrightarrow{k_a} C3 + C8$

9.) $proC8 + C3 \xrightarrow{k_a} C3 + C8$

10.) $F + R \xrightarrow{k_e} RF$

11.) $RF \xrightarrow{k_m} F + R$

8. (1 point) Which reaction in the FLIP system implements a positive feedback mechanism? Refer to the numbering scheme you used above.

Reaction 9 of the FLIP system is the reaction that implements the positive feedback mechanism. As active C3 activates proC8 into active C8, a positive feedback loop is created since C8 can activate additional proC3 into C3, which then in turn can activate more C8. Hence, a positive feedback loop system.

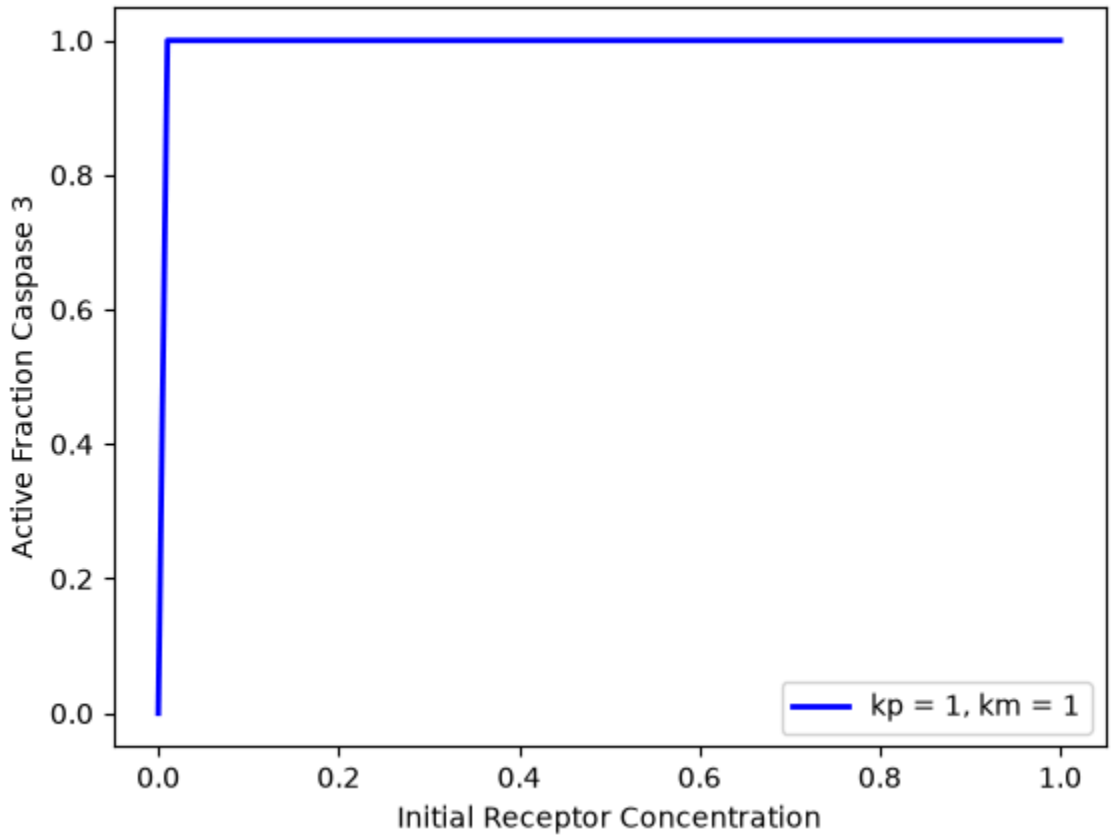
9. (5 points) In the template code, there is a section titled “Simple caspase model with feedback and Flip.” Write down the change equations for this model, and paste a screenshot below.

```
### ADD your change equations here
proC8prime = QC8 - delta*proC8 - k_ba*proC8*R - k_a*proC8*C3
C8prime = k_ba*proC8*R + k_a*proC8*C3 - delta*C8
proC3prime = QC3 - delta*proC3 - k_a*proC3*C8
C3prime = k_a*proC3*C8 - delta*C3
Rprime = -kp*R*F + km*RF
Fprime = -kp*R*F + km*RF
RFprime = kp*R*F - km*RF
###
```

We will once again explore steady state responses of the model in the section “STEADY-STATE responses of feedback + FLIP model”. In particular, we want to simulate how varying the concentration of initial active receptors affects the downstream activation of caspase 3.

Note we once again first define initial conditions and parameter values. All caspases exist only in their non-active state at the beginning of the simulation and the concentration of FLIP is set equal to that of the caspases. The rate of activation of pro-caspase 8 is slower to mimic the "basal" activity of proC8 (without feedback).

10. (3 points) Run this section of code and paste the resulting graph below.



11. (5 points) If complete activation of C3 leads to cell death, does the fate of the cell vary with receptor concentration?

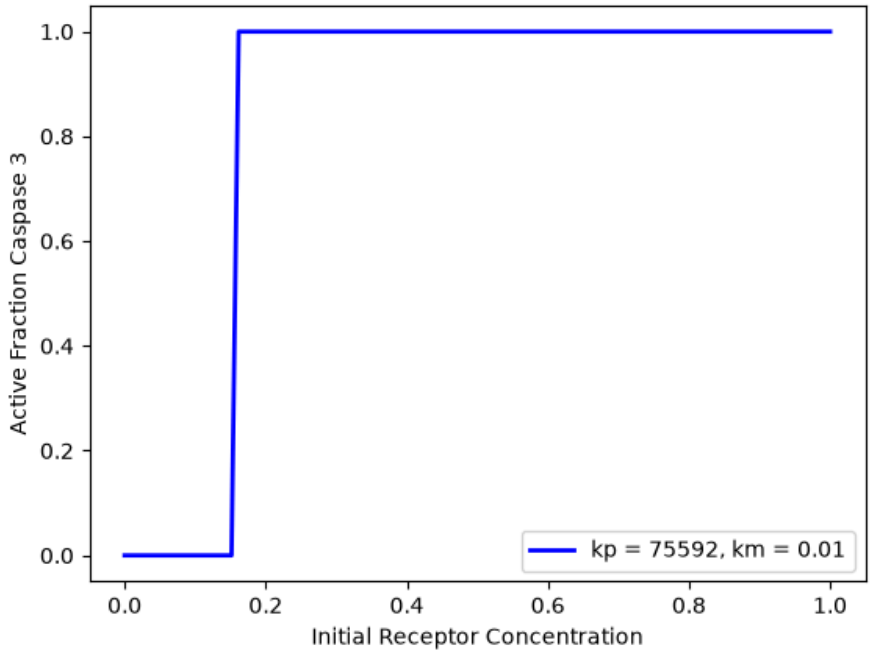
The cell fate does not significantly vary with the receptor concentration in this model. So as long as there is some active receptor present C3 becomes almost completely activated, which would lead to cell death. An increase of receptor concentration would not affect the outcome, because even very small concentrations of the receptor can be sufficient enough to trigger complete activation through the positive feedback loop. It is only when concentration is zero can the cell avoid C3 activation.

12. (3 points) Is this result biologically reasonable? Why or why not?

The model's result is not biologically reasonable as cells should not be able to undergo apoptosis in response to extremely minute levels of receptor activation. Random fluctuations or weak signals could activate a few receptors, but there should be a more quantifiable degree that once reached can activate the process of apoptosis. This more realistic system requires receptor activation to reach a certain threshold before C3 can become fully activated. This allows cells to avoid accidental apoptosis that could happen in the original model of question, while still being able to strongly respond to sufficient signals.

We are now going to modify some parameters to see how that changes the behavior of the model.

13. (5 points) Modify the values of kp and km and rerun "STEADY-STATE responses of feedback + FLIP model" until the plot generated differs from the previous result. When you generate different behavior, **paste that graph here**. Note that you may need to use some pretty extreme values of the parameters to achieve this goal. At these extreme values, you may encounter "Numerical artifacts," which will cause the graph to "jump up and down" in an unrealistic way. Try playing with the parameters to see if you can get a graph that does not do that!



14. (5 points) Explain why your modified parameter values generated a different behavior. Is this result more or less reasonable for than the previous result when considering cell fate and activated receptor concentration?

I increased kp to 75592, and set km to 0.01, theoretically making FLIP bind strongly to receptors relative to how quickly it dissociates. This would allow FLIP to sequester more receptors at low receptor concentrations, hence reducing the activation of C8 and the downstream positive feedback loop. My graph does show a threshold near 0.16 μM, where C3 activation changes sharply from zero to complete activation. This could be more biologically reasonable than the original result since cells should be able to tolerate weak or random receptor signals without undergoing apoptosis. But the extreme parameter values are causing numerical solver warnings, so the threshold may not be reliable.

Part 3: Irreversible Binding in the FLIP Caspase Model

Underneath the section where you plotted the steady state responses for the FLIP model, there is a section called “STEADY-STATE responses of feedback + irreversible FLIP model”. In this section, we will now assume that binding of FLIP to the receptor is irreversible. We will implement this using the same model as before (“caspase_FLIP_model”) and modifying a single parameter value.

15. (2 points) What parameter value do we change to simulate irreversible binding of FLIP to the receptor? Implement this change into the new section of the model code under the #parameters section, take a screenshot, then paste it below.

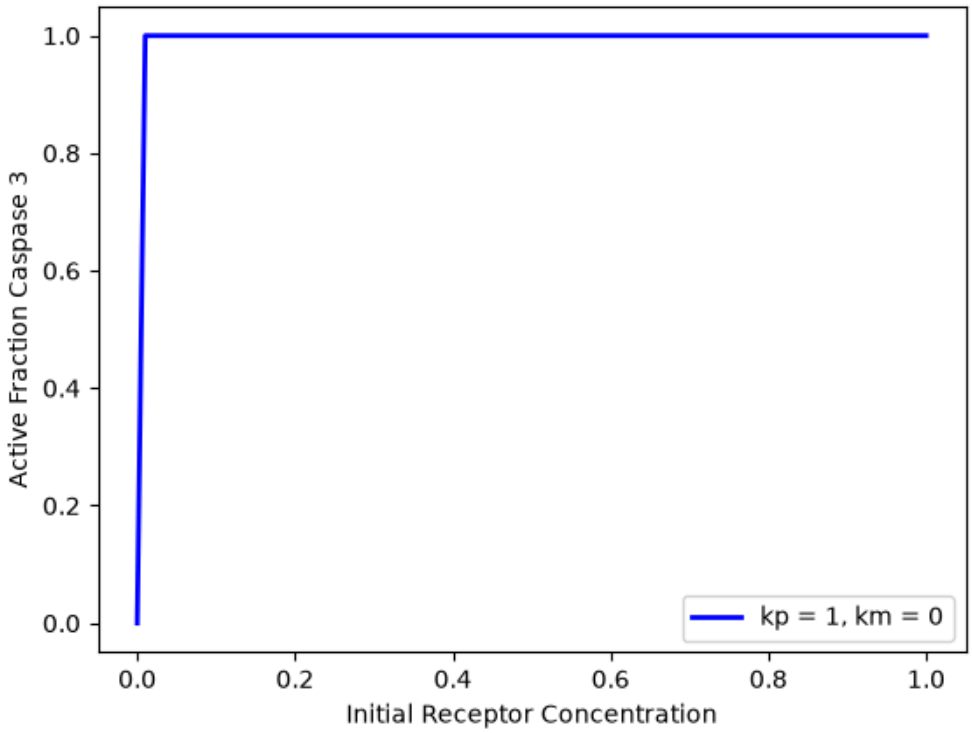
We changed the km value to 0 as shown below.

```
#parameters (You will have to change one of these to implement irreversible binding)
k_a = 1
k_ba = 0.001
kp = 1
km = 0
delta = 2e-4
QC8 = proC8_0*delta
QC3 = proC3_0*delta
```

The remaining parameters will remain unaltered to keep them in a biologically reasonable range.

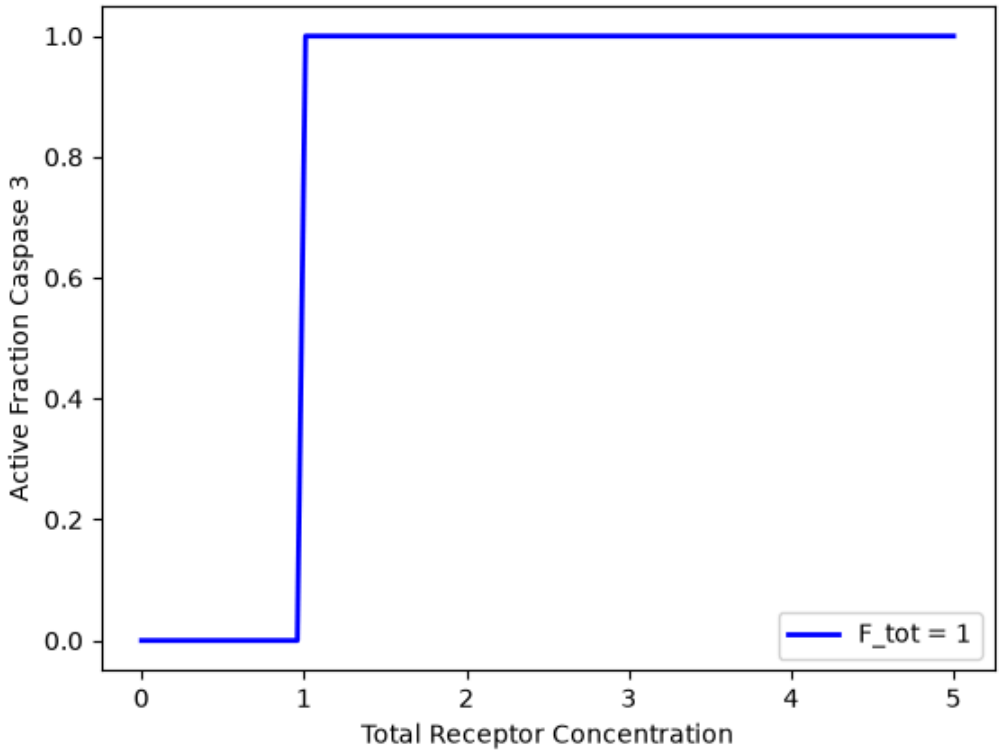
16. (5 points) Run your code with the modification you made in 15 and paste the plot of the steady-state result below. Does irreversible Flip binding alone generate biologically reasonable behavior?

The irreversible flip binding alone does not generate biologically reasonable behavior. The graph shows that C3 becomes fully activated at the first nonzero receptor concentration, so the same all or nothing response as seen before. Even a tiny amount of active receptor would trigger apoptosis.



We will now assume that FLIP is bound to the receptor prior to the start of the simulation. If the total amount of FLIP (F_tot) is greater than or equal to the total amount of receptors (R_tot), than all receptors are initialized to the receptor-FLIP complex (RF_0 = R_tot), R_0 = 0, and F_0 is equal to amount of FLIP in excess of R_tot. If F_tot is less than R_tot, than all FLIP is initialized to the receptor-FLIP complex (RF_0 = F_tot), F_0 = 0, and R_0 is equal to the amount of receptors in excess of FLIP.

17. (5 points) Implement these rules for the initial conditions, utilizing the if/then statement inside of the for loop in the section entitled “STEADY-STATE responses of feedback + irreversible FLIP model”. Run the section of code, then paste the resulting graph below.



18. (5 points) Explain in your own words what is happening in the model as the total receptor concentration surpasses the total FLIP concentration.

When the total receptor concentration is less than or equal to the total flip concentration, the receptors are all bound to FLIP and cannot activate C8. Since binding is irreversible, receptors stay blocked and C3 stays inactive which allows the cell to survive. Once receptor concentration becomes higher than FLIP concentration, free receptors are left over which can activate C8. Then C8 can be activated, which creates the positive feedback loop that causes more caspase activation, hence the sharp switch in the graph. The cell survives below the threshold but it will undergo apoptosis once the receptor concentration exceeds FLIP.

19. (3 points) In this model, how does FLIP concentration control C3 activation as a function of receptor concentration?

The FLIP concentration is what determines how much receptor activation is needed before C3 becomes fully activated. Since FLIP binds to receptors and prevents their activation of C8, increasing the amount of FLIP means more receptors are needed before any can remain free. This makes the activation threshold shift to a higher receptor concentration, hence the cell less likely to undergo apoptosis. Decreasing flip has the opposite effect, fewer receptors are blocked and C3 is allowed to be activated at lower receptor concentration. FLIP overall changes where the switch occurs.

To answer the following question, we need to establish some facts about parameter values for biological systems. Here are two simple principles to keep in mind:

1. Protein binding reactions in cells are **always reversible**.
2. Association rates and dissociation rates have very clear limits. Association rates like kp in this model essentially **never** get larger than 100 $\mu M^{-1} s^{-1}$. Dissociation rates like km can **never** get smaller than 0.001 s^{-1}.

20. (5 points) In questions 13 and 17, were you able to obtain the biologically reasonable case of a "bistable" cell fate at steady state as a function of increasing active receptor concentration (i.e. increasing R)? Were the assumptions you needed to make (parameters for kp and/or km, or reversible binding) reasonable according to the principles above? Based on your analysis, do you think the FLIP mechanism we discussed in class is a reasonable explanation for how cells avoid cell death at low concentrations of active receptor?

Both models were able to show something similar to a switch-like behavior, but only under conditions that were not very biologically reasonable. In question 13, kp was increased to 75592 for the best behavior, which realistically is much too high for binding rates. The extreme parameters also caused numerical warnings, which made the results less reliable. With realistic parameters, any amount of receptor activation caused complete C3 activation because of the positive feedback loop. Question 17 had a clear threshold appear when FLIP binding was irreversible, however protein binding is normally reversible so that

assumption as well may be unrealistic. Overall, these results suggest that FLIP alone may not be the complete model on how cell death is prevented at low receptor concentrations, and another mechanism may be needed for realistic activation thresholds.